ICT, it’s Indices, hardware and software

- What are ICT indices?
- Importance of ICT indices
- Various ICT indices: IDI, NRI, EGDI,GCI, NCSI

What are ICT Indices?

- To measure the strength of ICT development, deployment, usages - there are various standard indexes are used using various relevant parameters.

- Important ICT indices are composite measures developed by various organizations and compare a country's ICT development over time.

- These indices provide a standardized framework for assessing global digital divides by combining quantitative data on **ICT access, use, and skills, enabling governments and organizations to develop targeted policies for digital inclusion and economic growth.**

- These indices will help us to understand the current level of ICT development and deployment and finally, you may infer many points.

Purpose/Importance of ICT indices

1. **Monitoring Progress:**
   - Tracking changes in ICT development within and between countries.

2. **Identifying Gaps:**
   - Highlighting the extent of the digital divide.

3. **Informing Policy:**
   - Guiding the creation of policies and strategies to promote digital inclusion and leverage ICTs for economic and social development.

Various ICT indices

1. Networked Readiness Index (NRI)
2. ICT Development Index (IDI)
3. E-Government Development Index (EGDI)
4. Global Cybersecurity Index (GCI)
5. National Cyber Security Index (NCSI)

Networked Readiness Index (NRI)

- It is introduced by World Economic Forum (WEF). The WEF assesses the impact of ICT on the competitiveness of the nations of the world.
- The 3 components of NRI are:

  i) Environment for ICT (market, political and regulatory, infrastructural environment),
  ii) The Stakeholders (individuals, businesses and government) to use ICT, and
  iii) The usage of ICT.

  - In 2024, Bangladesh score is 43,56 (89th position out of 133 countries).

ICT Development Index (IDI)

- It is introduced by International Telecommunication Union (ITU). The IDI is a valuable tool for benchmarking the most important indicators for measuring information society or impact of ICT in the society.
- This index is dependent on eleven indicators, is grouped into 3-major categories:
  i) ICT readiness (Infrastructure, access) - reflecting the level of networked infrastructure and access to ICTs,
  ii) ICT Use (Intensity) - reflecting the level of use of ICTs in the society and
  iii) ICT capability (Skills) - reflecting the results/outcomes of more efficient and effective ICT use.
- **2025 IDI Score:** 64.9 (based on 2023 data), **Global Average:** 78, **Lower-Middle-Income Average:** 66

E-Government Development Index (EGDI)

- It is introduced by United Nations. The EGDI evaluates the impact of ICT on the political economy of a country.
- This index focuses on how the governments use ICT to deliver services to the people and opportunities for citizens to participate in decision making process.
- The EGDI consists of 3-components:
  i) Online service,
  ii) Technological infrastructure and
  iii) Human capital.
- In the **United Nations E-Government Development Index (EGDI) 2024**, Bangladesh ranked **100th out of 193 countries** with a score of 0.6570

Global Cybersecurity Index (GCI)

- It is introduced by International Telecommunication Union (ITU).
- The GCI results show overall improvement and strengthening of all 5 pillars of the cybersecurity agenda:

  i) Legal measures,
  ii) Technical measures,
  iii) Organizational measures,
  iv) Capacity development measures, and
  v) Cooperation measures.

- GCI 2024, Bangladesh ranked **53rd** out of 194 countries

National Cyber Security Index (NCSI)

- It is introduced by NCSI Project Team (e-Governance Academy Foundation), Estonia.
- The NCSI measures the preparedness of countries to prevent cyber threats and manage cyber incidents.
- The NCSI is also a database with publicly available evidence materials and a tool for national cyber security capacity building.
- The NCSI focuses on 4-measurable aspects of cyber security :
  - i) Legislation in force – Legal acts, regulations, orders, etc.,
  - ii) Established units – Existing organisations, departments, etc.,
  - iii) Cooperation formats – Committees, working groups, etc. and,
  - iv) Outcomes – Policies, exercises, technologies, websites, programs, etc.
- Bangladesh holds the 48th position in the National Cyber Security Index (NCSI) with a score of 66.67, according to 2021 data.